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Phase space bottlenecks and rates of no-barrier fragmentation reactions into polyatomic molecules

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An expression of the microcanonical unimolecular rate for an arbitrary transition state surface in phase space is derived and applied to fragmentation reactions into polyatomic molecules without potential barrier. The transition state which has a “point of no return” property in unimolecular dissociation is defined as an interfragment bottleneck in phase space. The fragmentation rate based on the interfragment bottleneck in phase space is compared with the rate based on the transition state defined in configuration space. The rate derived from the flux which crosses the interfragment bottleneck by intermode energy transfer is found to be smaller than the rate derived from the Rice–Ramsperger–Kassel–Marcus or phase space theory by an approximate factor $(s+r/2)|\tilde{W}|/E$, where E is the total energy and $|\tilde{W}|$ is the magnitude of the coupling energy between the reaction coordinate and the s -dimensional vibrational and r -dimensional rotational modes of the fragments. Phase space theory grossly overestimates the rate of fragmentation of small molecules with small $|\tilde{W}|$ in the high energy range, because the theory does not take into account the slow process of intramolecular energy redistribution. © 1996 American Institute of Physics. [S0021-9606(96)00245-0]

I. INTRODUCTION

A no-barrier thermal dissociation and association (D/A) reaction $\text{N}_2\text{O}_4 \rightleftharpoons 2\text{NO}_2$ on the ground state potential energy surface in the gas and liquid phases has been studied by classical molecular dynamics simulations including vibrational degrees of freedom.¹ Various microscopic reaction processes of N_2O_4 activated above the dissociation threshold have been studied in detail in the preceding paper.² Reactive energy and angular momentum transfers for fragmentation reactions were studied by dividing the motion of the system into the fragment's vibrational (V), rotational (R), and interfragment (T) modes, where the T mode is composed of interfragment translational (TT) and orbital (TL) modes as schematically shown in Fig. 1. In this report, we formulate a theoretical framework for the numerical results of reaction dynamics simulations in the preceding manuscript.

Transition state theory (TST), introduced by Eyring³ and by Evans and Polyani,⁴ provided the first theoretical attempt to determine absolute reaction rates. Two basic assumptions in deriving the TST rate expression are: (i) the transition state (TS) is a “point of no return,” which means that a molecular system that has crossed the TS in the direction toward product cannot return and reform a reactant; (ii) the internal energy redistribution in an activated molecule is much faster than unimolecular dissociation. In discussing the fragmentation of activated molecules in the absence of collisions, one usually starts with the potential energy surface as a function of a selected reaction coordinate. For bond breaking without barriers, there is no unique way to locate a TS at a specific position on the reaction coordinate. Two kinds of statistical theories for microcanonical rates have been proposed for unimolecular dissociation reactions; one is based on the TS defined in configuration space and the other is based on the TS defined more generally in phase space.

The Rice–Ramsperger–Kassel–Marcus (RRKM) theory for unimolecular reactions defines the TS in configuration space.^{5,6} If it is assumed that the TS is located at a particular value of the reaction coordinate, $R=R^\ddagger$, the microcanonical rate constant for unimolecular dissociation at a given energy E is given by

$$k_{\text{RRKM}}(E) = \frac{N_E(R^\ddagger)}{h\rho(E)}, \quad (1)$$

where $N_E(R^\ddagger)$ is the number of available energy levels for modes orthogonal to the reaction coordinate at the TS configuration with energy less than or equal to E , h is Planck's constant, and $\rho(E)$ is the density of states for the reactant at E . The orthogonal motions include transitional modes as well as conserved vibrations.^{5,6} Many theories which place the TS in configuration space are based on Eq. (1), but they differ in the location of the TS.

For bond breaking without barriers, the energy maximum is located at infinite separation of products. However, a barrier is formed at $R=R_l$ in the effective potential $V_{\text{eff}}(R)$ when the orbital angular momentum quantum number l of the separating fragments is large.² The barrier height $V_{\text{eff}}(R_l)$ is zero (i.e., $R_l=\infty$) for small l , and $V_{\text{eff}}(R_l)$ is slightly positive for large l . In phase space theory (PST),^{6–8} the TS is placed at the effective barrier, and the energy conservation law imposes the condition for dissociation:

$$E_T = E - E_V - E_R > V_{\text{eff}}(R_l) \geq 0, \quad (2)$$

where E_T, E_V, E_R represent the energies of the interfragment motion, vibrational, and rotational motions of fragments, respectively. In PST, $N_E(R^\ddagger)$ in Eq. (1) denotes the total number of these vibrational-rotational states which leave the energy E_T larger than the energy of the l dependent effective barrier.

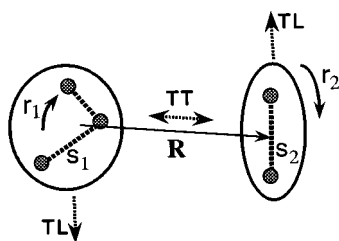


FIG. 1. Schematic representation of the degrees of freedom of fragmentation in the center of mass scheme. The fragment molecules 1 and 2 have s_1, s_2 dimensional vibrational modes and r_1, r_2 dimensional rotational modes. This is an example of $s_1=3, s_2=1, r_1=3, r_2=2$. $\mathbf{R}$ is the vector between centers of masses of fragment molecules, and TT and TL denote interfragment translational and orbital modes, respectively.

The PST predictions are in good agreement with the observed no-barrier photodissociation rate constants near threshold energy in the reactions $\text{NCNO} \rightarrow \text{CN} + \text{NO}$,⁸ and $\text{CH}_2\text{CO} \rightarrow \text{CH}_2 + \text{CO}$,^{9,10} but PST usually overestimates the rate constants at high energies. Consequently, the dynamical constraints that slow down the reaction were introduced in some models. In variational RRKM theory,^{11,12} the reaction coordinate has been factored out from other degrees of freedom, and the TS was placed at the value $R^\ddagger$ which minimizes the rate, i.e., $(\partial N_E(R)/\partial R)_{R=R^\ddagger} = 0$. The variational RRKM theory was found to give a good description of photodissociation rate constants of NCNO ⁸ and CH_2CO .^{9,10} However, the observed dissociation rates may be explained not by decreasing the number of accessible states at the TS, but by models which explicitly incorporate the rate of intrareactant energy redistribution.

A notable example of non-RRKM behavior is the vibrational predissociation of van der Waals molecules: for example,^{13,14}



where $\text{X} = \text{He}$ or Ar , and ν denotes the vibrational quantum number in the I_2 part. The variational RRKM theory for the fragmentation of the T-shaped HeI_2 , where the TS was placed variationally at $R^\ddagger = \infty$, was found to overestimate the rate by two orders of magnitude. The energy transfer from the I_2 bond to the van der Waals bond required for dissociation is much slower than the one assumed in the RRKM theory. The assumptions in RRKM theory that unimolecular complexes exist for $R < R^\ddagger$ and that the intramolecular energy redistribution is faster than dissociation are therefore not applicable. However, a statistical formulation which assumes a TS at the intermolecular bottleneck in phase space:

$$E_T = \frac{P^2}{2\mu} + V_T(R) = 0, \quad (4)$$

where P and R are respectively the canonical momentum and coordinate associated with the van der Waals degree of freedom, μ is the effective mass, and V_T is the effective interaction potential, is found to provide a much better estimate of the rate for fragmentation of HeI_2 .¹³

In our preceding molecular dynamics simulations of a no-barrier thermal dissociation and association reaction, various microscopic reaction processes of N_2O_4 activated above the dissociation threshold have been studied by assuming a potential for a loosely bound NO_2 pair under collisionless condition.² Energy E_T , which is defined as the sum of the potential and kinetic energies of interfragment motion, is found to play a key role in the D/A reaction; a reactant pair associates when $E_T < 0$ and the pair dissociates when $E_T > 0$. Association is found to occur when a pair of approaching fragments passes the effective barrier to association [$E_T > V_{\text{eff}}(R_l)$], and then E_T changes sign from positive to negative by TV (from T to V mode) and/or TR (from T to R) followed by RT (from R to T) energy transfers by an interfragment collision. Dissociation into fragments occurs by three steps: first, energy redistribution among $T-R-V$ modes at an inner turning point (ITP) of interfragment motion, where E_T becomes positive and the product vibrational distribution is determined; second, energy redistribution among $TT-TL-R$ modes at an outward region, where l is determined, E_T becomes larger than $V_{\text{eff}}(R_l)$, and the product rotational distribution is determined; and third, separation of fragments by transferring energy from TL to TT with constant E_T and l . These energy transfers were found to occur mainly centered at ITPs of the interfragment motion, and in rare cases, near outer turning points of interfragment motion by RT/TR energy transfers. The most appropriate TS of no return to product was identified as a phase space surface of $E_T = 0$, or more strictly $E_T = V_{\text{eff}}(R_l)$, where the reactive T mode acquires enough energy to surmount the l dependent effective barrier.

The present energy condition for the TS is completely in accord with the one of PST. On the other hand, the variational RRKM theory for no-barrier dissociation reactions was found not to be valid because $N_E(R^\ddagger)$ at the variationally determined TS contains a number of states which return to reactant.² In no-barrier reactions near threshold, the number of accessible states at the PST TS is small and the rate of dissociation is very low. Because the energy is randomized faster than dissociation in this case, the rate can be determined statistically by Eq. (1). As the energy increases, the dissociation rate increases and may become comparable to the rate of intrareactant energy redistribution. In this energy range, energy randomization among degrees of freedom may influence the rate, and the RRKM rate will give only an upper limit. Hence, it is important to develop a theory which explicitly incorporates the rate of intrareactant energy redistribution. In this paper, the fragmentation rate of polyatomic molecules for an arbitrary TS surface will be derived. The rate based on the phase space bottleneck TS will be compared with the rate based on the TS defined in configuration space. It will be shown that the rate depends strongly on the definition of the TS, and that the rate for the phase space bottleneck contains the effect of the intrareactant energy redistribution process.

II. FORMULATION OF TST FOR MULTIDIMENSIONAL UNIMOLECULAR DISSOCIATION

An expression of the general transition state unimolecular rate for an arbitrary surface in phase space was first derived by Gray *et al.* for the vibrational predissociation of van der Waals molecules HeI_2 and ArI_2 .^{13,14} We shall extend their formulation in a form which is applicable to fragmentation reactions into polyatomic molecules.

An approximate separation of variables into three groups in the center of mass scheme is appropriate for the approaching and separating fragments, where the fragment molecules 1 and 2 are assumed to have $s_i (i=1,2)$ dimensional vibrational modes and $r_i (i=1,2)$ dimensional rotational modes (Fig. 1). (i) The interfragment modes between centers of masses of fragment molecules, which are represented by $\mathbf{R}$ and its conjugate momentum $\mathbf{P}$. The reaction coordinate (the distance $R=|\mathbf{R}|$) is derived from a parent vibrational mode, while the angular part of $\mathbf{R}$ which describes the interfragment orbital motion is derived from parent rotation. (ii) The $r=r_1+r_2$ dimensional rotational modes of products ($r=4,5,6$ for diatom–diatom, diatom–polyatom, and polyatom–polyatom fragmentations, respectively), which are correlated with $r-1$ transitional modes and one parent rotational mode about the interfragment axis. (iii) The $s=s_1+s_2$ dimensional conserved modes, which are typically intrafragment vibrations and persist in the products. The total dimension of the system is $N+3$, where $N=r+s$.

Here we shall derive a general expression for unimolecular rates of a $N+3$ dimensional molecular system described by the classical Hamiltonian

$$H(\mathbf{P}, \mathbf{R}, \mathbf{p}, \mathbf{x}) = \frac{P^2}{2\mu} + \frac{\mathbf{L}^2}{2\mu R^2} + \sum_{i=1}^r \frac{l_i^2}{2I_i} + \sum_{j=r+1}^N \frac{p_j^2}{2\mu_j} + V(\mathbf{R}, \mathbf{x}), \quad (5)$$

where P denotes the conjugate momentum of R , $\mathbf{L}=\mathbf{R}\times\mathbf{P}$ denotes angular momentum of interfragment orbital motion, $\mathbf{p}=(p_1, p_2, \dots, p_N)$ and $\mathbf{x}=(x_1, x_2, \dots, x_N)$ denote canonical momenta and coordinates in which $(x_1, \dots, x_r)$ and $(x_{r+1}, \dots, x_N)$ are related to the orientational and internal coordinates of fragments, respectively. l_i is a component of angular momentum of each fragment with moment of inertia I_i about the inertial axis i , and μ_j is the effective mass for the conserved mode j . The V - R coupling exists through the dependence of the moment of inertia I_i on the vibrational coordinates of fragments, but it is neglected here as a zero-order approximation. The interfragment orbital angular momentum is determined in the final stage of dissociation by energy redistribution among TT – TL – R modes as we have seen in the last section. If the orbital angular momentum is assumed to be conserved in the approaching and separating processes of fragments, the dimension of the system can be reduced from $N+3$ to $N+1$ by substituting $\mathbf{L}^2=\hbar^2 l(l+1)$ in Eq. (5).

By this simplification, the TS in this system can be defined generally by a phase space surface

$$S(P, R, \mathbf{p}, \mathbf{x}) = 0, \quad (6)$$

where the convention $S<0$ implies that a phase point $(P, R, \mathbf{p}, \mathbf{x})$ is in the reactant side and $S>0$ implies a phase point is in the product side. The classical TST rate constant for a unimolecular reaction at an energy E is expressed as

$$k_S(E) = \frac{F_S(E)}{\rho_S(E)}, \quad (7)$$

where $F_S(E)$, the flux of phase points that crosses the transition state surface $S=0$ in the direction to product, is given by using $\dot{S}=dS/dt$ as

$$F_S(E) = h^{-(N+1)} \int dP dR d\mathbf{p} d\mathbf{x} \delta(S) \delta(E-H) \dot{S} \theta(\dot{S}), \quad (8)$$

and $\rho_S(E)$, the density of phase points of the reactant at the given E , is given by

$$\begin{aligned} \rho_S(E) &= h^{-(N+1)} \int dP dR d\mathbf{p} d\mathbf{x} \theta(-S) \delta(E-H) \\ &= h^{-(N+1)} \frac{d}{dE} \int dP dR d\mathbf{p} d\mathbf{x} \theta(-S) \theta(E-H). \end{aligned} \quad (9)$$

The reactant phase space in Eq. (10) is defined to be all the phase space within the TS surface through the step function θ , which is defined as $\theta(z \geq 0)=1$ and $\theta(z < 0)=0$.

Here we rewrite the potential energy $V(\mathbf{R}, \mathbf{x})$ in Eq. (5) according to an approximate separation of variables for a loosely interacting pair of fragments as

$$V(\mathbf{R}, \mathbf{x}) = V_T(R) + V_V(x_{r+1}, x_{r+2}, \dots, x_N) + W(\mathbf{R}, \mathbf{x}), \quad (11)$$

where $V_T(R)$ is the effective interaction potential in the reactive degree of freedom and $V_V(x_{r+1}, x_{r+2}, \dots, x_N)$ is the intramolecular potential energy of fragments. Then a zero-order approximation to the full Hamiltonian is given by

$$H_0 = h_T(P, R) + h_R + h_V \quad (12)$$

$$\begin{aligned} &= \left\{ \frac{P^2}{2\mu} + \frac{\hbar^2 l(l+1)}{2\mu R^2} + V_T(R) \right\} + \sum_{i=1}^r \frac{l_i^2}{2I_i} \\ &+ \left\{ \sum_{j=r+1}^N \frac{p_j^2}{2\mu_j} + V_V(x_{r+1}, x_{r+2}, \dots, x_N) \right\}, \end{aligned} \quad (13)$$

where h_T , h_R , and h_V represent the Hamiltonians for the reactive, rotational, and vibrational modes, respectively. The second and third terms of $h_T(P, R)$ constitute the effective potential $V_{\text{eff}}(R)$ for the reaction coordinate R , and the centrifugal barrier in V_{eff} is formed at $R=R_l$ when l is large. $W(\mathbf{R}, \mathbf{x})=V(\mathbf{R}, \mathbf{x})-V_T-V_V$ gives the coupling among T , R , and V modes, and as the result, a molecule dissociates. This term also gives rise to internal energy redistribution in the RRKM picture of fragmentation.

As we have seen in the last section, the TS which has a point of no return property to product in no-barrier fragmentation is not based on a simple configuration space condition such as

$$S(R) = R - R^\ddagger = 0, \quad (14)$$

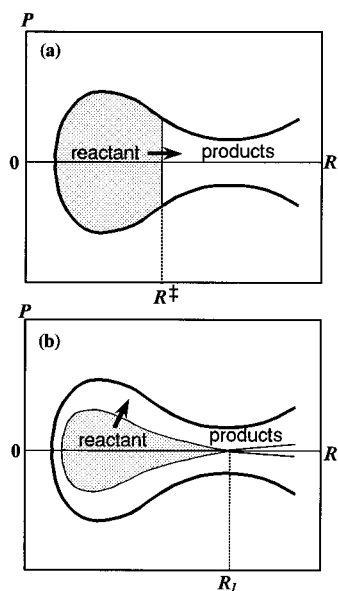


FIG. 2. Projections of transition states on the two dimensional (P, R) plane: (a) TS is defined in configuration space, $R - R^\ddagger = 0$; and (b) TS is defined in phase space, $h_T(P, R) - V_{\text{eff}}(R_l) = 0$. Fluxes across the transition states are shown by arrows.

as in RRKM theory, but rather is based on the phase space condition which depends on l

$$S(P, R) = h_T(P, R) - V_{\text{eff}}(R_l) = 0, \quad (15)$$

where $V_{\text{eff}}(R_l)$ is the l dependent barrier height, which is slightly positive. This expression defines $(2N+1)$ dimensional surface in $(2N+2)$ dimensional phase space. Projections of transition states [Eqs. (14) and (15)] on the two dimensional (P, R) plane are schematically shown in Fig. 2. Fluxes which cross the transition states are shown by arrows.

The standard RRKM theory rate expression for the transition state surface given in Eq. (14) can be calculated from Eqs. (7)–(10) as

$$\begin{aligned} k_{\text{RRKM}}(E) &= h^{-(N+1)} \rho_S(E)^{-1} \int dP dR d\mathbf{p} d\mathbf{x} \\ &\quad \times \delta(R - R^\ddagger) \frac{P}{\mu} \theta(P) \delta(E - H) \\ &= h^{-(N+1)} \rho_S(E)^{-1} \int d\mathbf{p} d\mathbf{x} \\ &\quad \times \theta(E - V_T(R^\ddagger) - h_R - h_V), \end{aligned} \quad (16)$$

which can be reexpressed into the more familiar RRKM form given in Eq. (1) by using the relation

$$N_E(R^\ddagger) = h^{-N} \int d\mathbf{p} d\mathbf{x} \theta(E - V_T(R^\ddagger) - h_R - h_V), \quad (18)$$

where the density of states for the reactant at E is given by

$$\begin{aligned} \rho_S(E) &= h^{-(N+1)} \frac{d}{dE} \int dP dR d\mathbf{p} d\mathbf{x} \theta(R^\ddagger - R) \\ &\quad \times \theta(E - H). \end{aligned} \quad (19)$$

On the other hand, the rate k_{PS} for the TS defined by the phase space bottleneck can be obtained by inserting Eq. (15) into Eq. (7) as

$$\begin{aligned} k_{\text{PS}}(E) &= h^{-(N+1)} \rho_S(E)^{-1} \int dP dR d\mathbf{p} d\mathbf{x} \\ &\quad \times \delta(h_T - V_{\text{eff}}(R_l)) \dot{h}_T \theta(\dot{h}_T) \delta(E - H), \end{aligned} \quad (20)$$

where the time derivative of h_T may be evaluated by the Poisson–Bracket relation

$$\dot{h}_T = \{h_T, H\} = \frac{\partial h_T}{\partial R} \frac{\partial H}{\partial P} - \frac{\partial h_T}{\partial P} \frac{\partial H}{\partial R} \quad (21)$$

$$= -\frac{P}{\mu} \frac{\partial W}{\partial R}. \quad (22)$$

Equation (20) then becomes

$$\begin{aligned} k_{\text{PS}}(E) &= h^{-(N+1)} \rho_S(E)^{-1} \int dP dR d\mathbf{p} d\mathbf{x} \\ &\quad \times \delta(h_T - V_{\text{eff}}(R_l)) \theta(P) \frac{P}{\mu} \left| \frac{\partial W}{\partial R} \right| \delta(E - H) \\ &= h^{-(N+1)} \rho_S(E)^{-1} \int dR d\mathbf{p} d\mathbf{x} \left| \frac{\partial W}{\partial R} \right| \\ &\quad \times \delta(E - V_{\text{eff}}(R_l) - h_R - h_V), \end{aligned} \quad (23)$$

where the coupling energy W is assumed to be small in comparison with h_T , h_R , and h_V . If the motion along the reaction coordinate is slow in comparison with the one of other vibrational modes of the parent molecule, $W(\mathbf{R}, \mathbf{x})$ can approximately be replaced by an average $\bar{W}(R)$ over phases of the conserved modes and over angles of two fragments. Then the integral over R gives the general unimolecular dissociation rate for the TS in phase space

$$\begin{aligned} k_{\text{PS}}(E) &= h^{-(N+1)} \rho_S(E)^{-1} \left[\int \left| \frac{\partial \bar{W}(R)}{\partial R} \right| dR \right] \int d\mathbf{p} d\mathbf{x} \\ &\quad \times \delta(E - V_{\text{eff}}(R_l) - h_R - h_V) \\ &= h^{-(N+1)} \rho_S(E)^{-1} \left[\int \left| \frac{\partial \bar{W}(R)}{\partial R} \right| dR \right] \frac{d}{dE} \\ &\quad \times \int d\mathbf{p} d\mathbf{x} \theta(E - V_{\text{eff}}(R_l) - h_R - h_V). \end{aligned} \quad (24)$$

The average phase approximation is useful for an order estimation of thermal dissociation rates, because the weakest bond breaks and the motion along the reaction coordinate is considered to be slow. Since $W(\mathbf{R}, \mathbf{x}) = 0$ in the limit of infinite separation of fragments

$$\int_{R_{\text{ITP}}}^{\infty} \left| \frac{d\bar{W}}{dR} \right| dR \approx |\bar{W}(R_{\text{ITP}})|, \quad (27)$$

where R_{ITP} is the interfragment distance around inner turning points. This approximation represents that the dominant flux across the TS surface occurs around inner turning points of interfragment motion. This is in accord with the observation

in our preceding reaction dynamics simulations² that transitions among T , R , and V modes which cross the TS surface $E_T \approx 0$ occur mostly around inner turning points. Equation (26) can then be reexpressed as

$$k_{\text{PS}}(E) = \frac{|\tilde{W}(R_{\text{ITP}})|}{h\rho_S(E)} \frac{dN_E(R_l)}{dE}, \quad (28)$$

where

$$N_E(R_l) = h^{-N} \int d\mathbf{p} d\mathbf{x} \theta(E - V_{\text{eff}}(R_l) - h_R - h_V) \quad (29)$$

$$\rho_S(E) = h^{-(N+1)} \frac{d}{dE} \int dP dR d\mathbf{p} d\mathbf{x} \times \theta(V_{\text{eff}}(R_l) - h_T) \theta(E - H), \quad (30)$$

$N_E(R_l)$ is the sum of states at the effective barrier R_l , and $\rho_S(E)$ is the density of states for the reactant. An approximation for the rate by Eq. (28) is found to give a reasonable agreement with the more accurate rate calculated by Eq. (24) for the T-shaped HeI_2 fragmentation.¹⁴

The ratio of the rate based on the phase space bottleneck to the RRKM rate at the TS of PST ($R^\ddagger = R_l$) can be calculated from Eqs. (28) and (1) as

$$\frac{k_{\text{PS}}(E)}{k_{\text{RRKM}}(E)} = \frac{|\tilde{W}(R_{\text{ITP}})|}{N_E(R_l)} \frac{dN_E(R_l)}{dE}, \quad (31)$$

where the difference in $\rho_S(E)$ due to different definitions of the reactant phase space [Eqs. (19) and (30)] is neglected. In order to make an order estimation for the ratio, we have to use a model for $N_E(R_l)$, the number of available energy levels for modes orthogonal to the reaction coordinate at R_l with energy less than or equal to E .

The number of states for $s = s_1 + s_2$ harmonic oscillators of fragments of frequencies $\nu_i (i = 1, \dots, s)$ with energy less than or equal to E has been obtained based on statistical mechanics as (Ref. 5, p. 528)

$$N_{\text{vib}}(E) = \frac{E^s}{s! \prod_{i=1}^s h \nu_i}. \quad (32)$$

The number of rotational states for $r = r_1 + r_2$ rotational degrees of freedom with energy less than or equal to E is obtained as (see appendix)

$$N_{\text{rot}}(E) = B(r_1, r_2) E^{r/2}, \quad (33)$$

where $B(r_1, r_2)$ are the constants which depend on the moments of inertia of the two fragments.

The number of vibrational-rotational states of separating fragments having $s = s_1 + s_2$ vibrational degrees of freedom and $r = r_1 + r_2$ rotational degrees of freedom with energy less than or equal to E can be calculated by using Eqs. (32) and (33) as

$$N_E = \int_0^E N_{\text{vib}}(E') \frac{dN_{\text{rot}}(E - E')}{dE} dE' \quad (34)$$

$$= \frac{B(r_1, r_2)}{s! \prod_{i=1}^s h \nu_i} \int_0^E E'^s \frac{r}{2} (E - E')^{r/2-1} dE' \quad (35)$$

$$= \frac{B(r_1, r_2)}{\prod_{i=1}^s h \nu_i} \frac{\Gamma(r/2 + 1)}{\Gamma(s + r/2 + 1)} E^{s+r/2}, \quad (36)$$

where Γ represents the gamma function $\Gamma(t) = \int_0^\infty e^{-x} x^{t-1} dx$. We then get

$$\frac{dN_E}{dE} = (s + r/2) \frac{B(r_1, r_2)}{\prod_{i=1}^s h \nu_i} \frac{\Gamma(r/2 + 1)}{\Gamma(s + r/2 + 1)} E^{s+r/2-1}, \quad (37)$$

which yields the ratio $(dN_E/dE)/N_E = (s + r/2)/E$. We finally obtain an approximate ratio of k_{PS} to k_{RRKM} as

$$k_{\text{PS}}/k_{\text{RRKM}} \approx (s + r/2) |\tilde{W}|/E, \quad (38)$$

where $|\tilde{W}|$ is the abbreviation of $|\tilde{W}(R_{\text{ITP}})|$. Because the magnitude of intermode coupling energy $|\tilde{W}|$ between the reaction coordinate and the rotational and vibrational modes of fragments is small as compared with total energy E , k_{PS} is found to be smaller than k_{RRKM} by an approximate factor $(s + r/2) |\tilde{W}|/E \ll 1$, where E should be replaced by $[E - V_{\text{eff}}(R_l)]$ if the effect of the l dependent potential barrier is not negligible.

III. DISCUSSIONS

Two kinds of statistical rates for unimolecular dissociation; one is based on the TS in configuration space and the other on the TS in phase space, have been reexpressed as the general transition state unimolecular rate for an arbitrary surface in phase space. The fragmentation rate derived from the flux which crosses the phase space bottleneck is found to be smaller than the rate from RRKM or PST by an approximate factor given in Eq. (38).

This equation shows three interesting trends. (i) The ratio $k_{\text{PS}}/k_{\text{RRKM}}$ increases (k_{PS} approaches k_{RRKM}) as energy E decreases and approaches the threshold energy. This is because the rate of fragmentation is very low and hence energy randomization occurs faster than dissociation. The dissociation rate is determined by the RRKM theory in this case. As energy E increases above threshold for reaction, the dissociation rate increases and the rate is influenced by the transition probability from stable reactant state to the TS states which directly lead to products. k_{RRKM} gives only an upper limit in this case. (ii) The rate k_{PS} increases as the intermode coupling energy W increases. The RRKM picture is a good model for the rapid energy randomization limit of $|\tilde{W}| \rightarrow E$. (iii) The rate k_{PS} is expected to be small for small size molecules of small vibrational degrees of freedom s , and increases proportionally to s in large molecules. Hence k_{PS} approaches k_{RRKM} as the molecular size increases.

The marked difference between k_{PS} and k_{RRKM} arises from different mechanisms of crossing the TS, and it is illustrated in Fig. 2. The flux across the phase space bottleneck

occurs by intrareactant energy transfer from the R and V modes to the T mode, and the dissociation is a part of energy redistribution processes in which the trajectory crosses the TS surface. On the contrary, no energy transfer to the reactive mode is needed in RRKM theory, and the momentum to cross the TS at $R=R^\ddagger$ is given by intrareactant energy redistribution, which is assumed to occur faster than dissociation. Intrareactant energy randomization occurs by the intermode coupling potential W , which is the potential responsible also for the flux across the phase space bottleneck. An order estimate of the ratio, $k_{\text{PS}}/k_{\text{RRKM}} \approx (s+r/2)|\dot{W}|/E \ll 1$, shows that intrareactant energy redistribution is slow as compared with dissociation, and hence energy randomization among degrees of freedom influences the rate. This is especially true for small molecules of small intermode coupling energy $|\dot{W}|$ with high energy E . The experimental and theoretical studies show that intramolecular energy redistribution is not complete in the overtone initiated dissociation reaction $\text{HOOH} \rightarrow 2\text{OH}$.^{15,16} Phase space theory predicts the observed no-barrier photodissociation rate constants of NCNO and CH_2CO near threshold energy but usually overestimates the rate constants at high energies.^{8–10} This is presumably because internal energy redistribution is slow and influences the rate.

Vibrational predissociation of van der Waals molecules is another example of slow internal energy randomization.^{13,14} The line width of the fluorescence excitation spectrum of ArI_2 shows strong dependence on the specific energy level prepared by the exciting light.¹⁷ The line-width is attributed to the electronic and vibrational predissociation rates. Vibrational predissociation rates calculated by classical trajectory simulations and quantum mechanics based on Fermi golden rule approach are both very nonstatistical; they do not increase monotonically, but oscillate with energy.^{13,14} An inverted parabola pattern appears in the dissociation rate from a separate ν manifold with increasing n , where ν denotes the vibrational quantum number in the I_2 part and n is the one in the van der Waals degree of freedom. Serious violations of rapid intramolecular energy exchange assumption seem to occur. However, the inverted parabola pattern may tend to wash out if sufficient state selection is not achieved in the initial excitation step. This seems to be the case in Ref. 17, and the pattern is not observed. Hence, if an ensemble of phase points that fills evenly the volume within the TS surface is selected as initial conditions of classical trajectory simulations or quantum mechanical calculations, the decay rate is expected to correlate with the statistical rate k_{PS} .

The predictions of RRKM theory for the fragmentation rates of van der Waals molecules are always too high, while the rates obtained based on the phase space bottleneck TS [Eq. (24)] are in good agreement with experiment and the classical and quantum rate constants averaged over the van der Waals manifold of states.^{13,14} This agreement is an example of how statistical theories can provide useful information even when the fundamental assumption of rapid intrareactant energy redistribution is not always satisfied. Quantitative understanding of the fragmentation process re-

quires rigorous quantum mechanical calculations. However, a careful study of the condition under which statistical theories describe some aspects of the dynamics is useful for obtaining a physical picture of the fragmentation processes of polyatomic molecules.

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APPENDIX: THE NUMBER OF ROTATIONAL STATES

The number of rotational states with energy less than or equal to E has been obtained for a two-dimensional linear molecule with moment of inertia I as⁵

$$N_{\text{rot},2}(E) = B_2 I E, \quad B_2 = \frac{8\pi^2}{h^2}. \quad (\text{A1})$$

For a three-dimensional nonlinear molecule with principal moments of inertia I_a, I_b, I_c , and angular velocities $\omega_a, \omega_b, \omega_c$, the rotational Hamiltonian is

$$H_{\text{rot},3} = \frac{1}{2}(I_a \omega_a^2 + I_b \omega_b^2 + I_c \omega_c^2), \quad (\text{A2})$$

and the number of rotational states with energy less than or equal to E can be calculated as

$$N_{\text{rot},3}(E) = h^{-3} \int_{H_{\text{rot}}=0}^{H_{\text{rot}}=E} dp_\alpha dp_\beta dp_\gamma d\alpha d\beta d\gamma \quad (\text{A3})$$

$$= h^{-3} I_a I_b I_c \int_{H_{\text{rot}}=0}^{H_{\text{rot}}=E} d\omega_a d\omega_b d\omega_c \int_0^{2\pi} d\alpha \\ \times \int_0^\pi \sin \beta d\beta \int_0^{2\pi} d\gamma, \quad (\text{A4})$$

where (α, β, γ) and $(p_\alpha, p_\beta, p_\gamma)$ are Eulerian angles and their conjugate momenta, respectively. In deriving Eq. (A4), variables are transformed from $(p_\alpha, p_\beta, p_\gamma)$ to $(\omega_a, \omega_b, \omega_c)$ using the Jacobian relation $dp_\alpha dp_\beta dp_\gamma = I_a I_b I_c \sin \beta d\omega_a d\omega_b d\omega_c$.¹⁸ The integral for the angular part gives $8\pi^2$ and the integral for angular velocities corresponds to the phase space volume of three-dimensional ellipsoid with semiaxes $(2E/I_a)^{1/2}, (2E/I_b)^{1/2}, (2E/I_c)^{1/2}$, which is $4\pi(2E)^{3/2}/3(I_a I_b I_c)^{1/2}$. The formula for the number of rotational states for a three-dimensional rigid rotor is then given by

$$N_{\text{rot},3}(E) = B_3 (I_a I_b I_c)^{1/2} E^{3/2}, \quad (\text{A5})$$

where $B_3 = 64\sqrt{2}\pi^3/3$. The number of rotational states including $r=r_1+r_2$ rotational degrees of freedom of fragments with energy less than or equal to E can be calculated using $N_{\text{rot},r_i}(E)$ for the two fragments $i=1,2$ as

$$N_{\text{rot}}(E) = \int_0^E N_{\text{rot},r_1}(E') \frac{dN_{\text{rot},r_2}(E-E')}{dE} dE' \quad (\text{A6})$$

$$= B(r_1, r_2) E^{r/2}, \quad (\text{A7})$$

where

$$B(2,2) = \frac{1}{2} B_2^2 I_1 I_2, \quad B(2,3) = \frac{2}{5} B_2 B_3 I_1 (I_{2a} I_{2b} I_{2c})^{1/2},$$

$$B(3,3) = \frac{3\pi}{32} B_3^2 (I_{1a} I_{1b} I_{1c} I_{2a} I_{2b} I_{2c})^{1/2},$$

I_i is the moment of inertia of linear rotor i , and I_{ja}, I_{jb}, I_{jc} are the principal moments of inertia of nonlinear molecule j .

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